



Security Analysis for Blind-Spot Collision Warning / Blind-Spot Collision -Avoidance Assist and Safe Exit Assist System

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Foundations of Cyber Security 2025

COMSM0118_2025_TB-1

October 02, 2025

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1 Introduction

The Blind-Spot Collision Warning (BCW) and Blind-Spot Collision-Avoidance Assist (BCA) systems, and the Safe Exit Assist (SEA) feature, are advanced driver assistance systems (ADAS) made to monitor & minimize the risks associated with blind-spot collisions and enhance vehicle safety.

1.1 Blind-Spot Collision Warning (BCW)

- BCW monitors the blind-spot via rear bumper radar sensors. The system warns the driver when another vehicle is near the blind spot.
- BCW's primary alert system has two stages. The first stage is a continuous warning light on the outer side mirror and head-up (if equipped).
- The second stage is a higher priority alert which includes a warning chime and a blinking light on the mirror/display.
- This alert is activated when the turn signal is detected, and a vehicle is detected in the blind spot.
- BCW activates when the vehicle is above 20 mph (30 km/h).

1.2 Blind-Spot Collision-Avoidance Assist (BCA)

- BCA uses the front upper-windshield camera and radar sensors for the sides and rear.
- This system may detect an imminent collision and activate the Electronic Stability Control (ESC) system to apply brakes softly on the wheel opposite to the potential impact.
- This is meant to assist in collision avoidance and lessen aggressive collision damage.
- The BCA system is enabled at vehicle speeds between 40 and 110 mph (60 and 180 km/h), driving with both detected lane markings and an approaching vehicle.

1.3 Safe Exit Assist (SEA)

- SEA is an operational unit that gets instructions from the BCW core logical controller.
- This system is built to perform safety-critical functions, including collision avoidance and lane change assistance.
- The Safe Exit Assist feature disallows the unlocking of the door when a car approaches from the back.

1.4 Offline Model Assumption

This analysis mainly focuses on interaction between User ↔ BCW & BCA ↔ Sensor, untouched the Blue Link system or the cloud. Thus, our DFD or attack trees should be considered as offline models, without interaction from the internet.

2 Data Flow Diagram

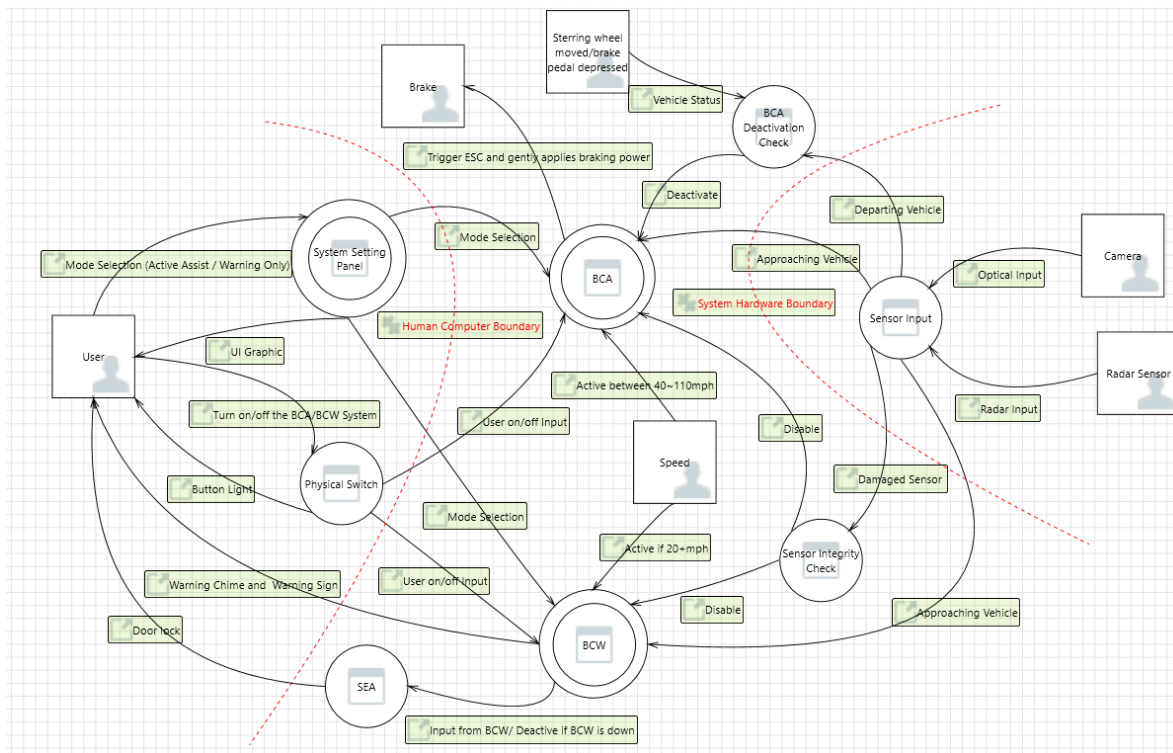


Figure 1: Data Flow Diagram

Based on the diagram above, the BCW & BCA system is divided into two boundaries, the Human-Computer and System-Hardware. Boundaries act as a “firewall” in the DFD. The first boundary lies between user input and BCW & BCA system, and the second crosses between system and sensor inputs, because user/sensor inputs are prone to be compromised.

The user input mainly comes from two interfaces. The system setting panel provides a graphical user interface (GUI), allowing the driver to select the operational mode (Active Assist / Warning Only). Alternatively, a physical switch enables the driver to turn the whole BCW & BCA system on/off.

System operation is divided between two functional units: BCW and BCA. BCW is activated when vehicle speed exceeds 20 mph. It processes both optical data and radar signals to detect approaching vehicles. When a potential hazard is identified, BCW triggers both an audio warning and a visual alert on the corresponding outer side-view mirror. If sensors are covered or damaged, BCW & BCA will be deactivated, notifying users with specific signs as well.

Instead of warning users, the BCA subsystem enhances driver safety by proactively intervening in the driving process. BCA is activated if Active Assist mode selected and vehicles in speed range 40-110 mph. In potential collision cases, BCA applies gentle braking. BCA is automatically deactivated under specific conditions such as an abrupt change in vehicle course, a sharp turn of the steering wheel, or depression of the brake pedal.

Furthermore, the system interfaces with the Safe Exit Assist (SEA) feature. With data input from BCW, SEA keeps door locked if approaching vehicles detected. Furthermore, the system interfaces with the Safe Exit Assist (SEA) feature, preventing the door from being opened and mitigating potential side-impact risks.

3 STRIDE analysis for two system

Before we draft a certain analysis on threat, one thing must do is to define the trusted boundaries. In the current model, there are mainly two of them: 1) Human - Computer Interaction and 2) System - Hardware Interaction.

3.1 Human Computer Interaction

S: An attacker could fake the driver input for turning on/off the BCW & BCA system.

T: An attacker could modify the BCW & BCA status display image from the system panel to trick the driver into believing that the system function is enabled/disabled.

R: The attacker deletes the HCI interaction and diagnostic logs, making their actions impossible to trace later.

I: HCI leaks user's driving preferences, blind spot trigger logs, or location-related data to malicious entity or to the cloud.

D: By exhausting HCI resources or triggering abnormal conditions (large number of requests/malicious input), the HMI cannot display or respond to critical alarms in a timely manner.

E: An attacker can gain access to engineering mode and modify security settings by cracking diagnostic authentication or exploiting HMI vulnerabilities.

3.2 System Hardware Interaction

S: The attacker forges radar echoes or optical imageries to mislead the signal processor, thus overwhelming BCA/BCW with false positives perception information (or negative positive), badly influencing the driver.

T: Tampering with the behavior and parameters of sensors through physical access (such as debug ports) or firmware replacement.

R: Attackers can tamper with or delete sensor and diagnostic logs at the hardware layer to conceal their interference with the system.

I: Sensor data or diagnostic information (such as location information and fault logs) is read or sent out through the gateway without authorization.

D: Sensors are easily interrupted by any kind of interference such as electromagnetic waves, jamming, and physical demolition blocking, which will further disable BCA/BCW.

E: Exploiting firmware vulnerabilities or weaknesses in the system to move laterally and gain control of the Secure Execution Unit (SEA) or other components.

4 STRIDE analysis with three threats in details

4.1 Human Computer Interaction

4.1.1 Elevation of Privilege

The core threat of this attack chain is privilege escalation. The attacker, starting with a role with limited access to a low-privilege physical interface (OBD-II), gradually elevates themselves to the highest level of privilege over the vehicle's critical electronic control units. This process involves three key stages: First, by cracking diagnostic security authentication (e.g., by brute-forcing keys or reverse engineering firmware), the attacker elevates privileges from a standard communication level to the "authorized diagnostician" level, unlocking reprogramming capabilities. Subsequently, by abusing the device control service to enter a programming session, their privileges are further elevated to the "system developer" level, gaining the ability to modify the firmware. Finally, by uploading and flashing malicious HMI firmware, the attacker successfully establishes a persistent and highly privileged backdoor in the HMI system, achieving the ultimate privilege escalation from an external attacker to an internal controller, enabling them to arbitrarily tamper with the security features.

4.2 System Hardware Interaction

4.2.1 Spoofing

One of the significant threats to the System - Hardware interaction is spoofing, in which a threat agent intercepted the sensor input and replaced with the well-crafted malicious inputs to the system, such as radar echoes and optical imagery. Since the BCW & BCA system highly relies on physical signal input, spoofing can cause harmful outcomes by affecting both BCW and BCA systems. For example, the attacker can manage to mask real approaching vehicles, but the system fails to warn or avoid the user. On the contrary, a malfunctioning system might detect fake objects on the road, triggering emergency braking. Moreover, fake objects generated could possibly affect the SEA system who relied on BCW input, and the door will be locked even in safe zone.

The BCW & BCA system expects that the physical layer returns true and trusted signal data. Therefore, the attack chain typically requires:

- (1) the capability to emit electromagnetic signals in certain bands or optical imaginary
- (2) the knowledge of physical sensor deployed on Santa Fe vehicles, such as wave bands
- (3) the basic understanding about BCW & BCA input-processing algorithm

The cyber kill chain will be elaborated in the section about Attack Trees.

4.2.2 Denial Of Service

Denial of Service has a significant impact on the System resilience, as an attacker or environmental situation blocks the radar or camera sensors to provide usable data or worse still, disables the system. BCW/BCA and SEA require constant sensor feed, and therefore, blocking, jamming, or damaging these sensors can prevent the functionality. As an example, Safe Exit Assist does not unlock a door when a vehicle is coming behind it, but when sensors cannot operate correctly due to jamming or obstructions, the Assistance can fail and leave the door open, which makes the situation dangerous. The system manual also cautions that these functions may be lowered or disabled by sensor pollution, fog or malfunction. These are practically DoS situations and may be used by the attackers.

Availability is the primary security principle that a successful Dos breaks. The system anticipates sensor information to make safe decisions, in case of loss of availability the system might not alert users and this poses a greater risk to safety.

5 Attack trees

5.1 Elevation of Privilege

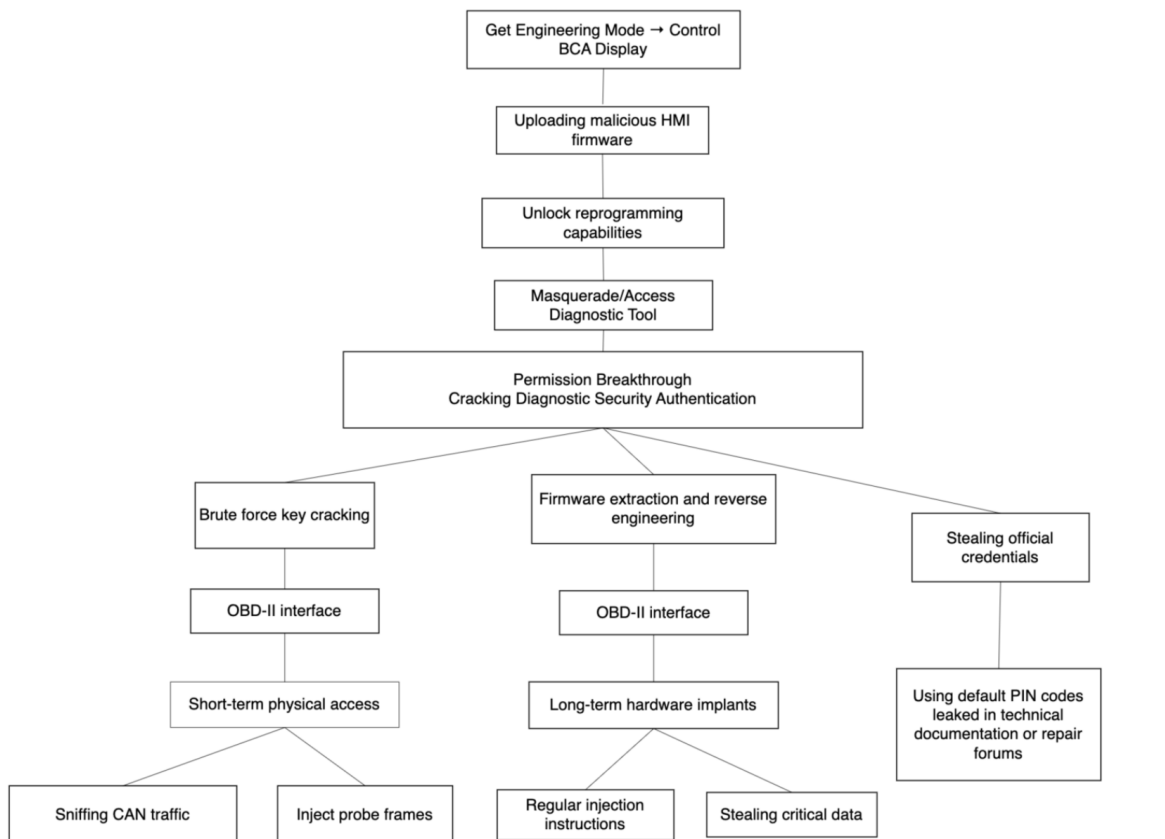


Figure 2: Attack Tree for Elevation of Privilege

Based on the attack tree diagram, the attack begins with physical access to the OBD-II interface, where adversaries either conduct short-term operations—such as sniffing CAN traffic and injecting probe frames—or deploy long-term hardware implants capable of periodic

command injection and critical data exfiltration [1, 2]. The core phase, labeled 'Permission Breakthrough,' focuses on cracking diagnostic security authentication. Attackers achieve this through brute-force key attacks, firmware extraction and reverse engineering, or by exploiting leaked credentials such as default PINs from technical documentation or repair forums.

Once authenticated, the attacker disguises as a legitimate diagnostic tool to enter a programming session, unlocking the ECU's reprogramming function. This enables the uploading of malicious HMI firmware, leading to the primary goal: gaining elevated privileges in the HMI engineering mode to control the Blind-Spot Collision Avoidance (BCA) system status display. Such remote exploitation and system manipulation demonstrates an escalation of capabilities similar to those demonstrated in real-world cases [3]. Finally, the attacker clears system logs to erase traces of the attack, ensuring persistent undetected manipulation of vehicle safety indicators.

5.2 Spoofing



Figure 3: Attack Tree for Spoofing

For the spoofing attacks happened in System – Hardware zone, attack vector could divide in two main branches: radar and camera.

5.2.1 Compromise of Radar signals

At the lowest level the attacker first needs to have basic understanding about the radar data envelop, such as frequency band and pulse timing etc. Additionally, attacker should have set up an equipment capable of emitting energy into the radar sensing zone. For example, a transmitter able to produce signals in the radar band. At the same time, the attacker is

required to have the opportunity to install the equipment on the vehicles to start the spoofing attack for a specific car. To mass influence multiple vehicles within certain areas, a high-power radar transmitter is required, which however beyond the scope of this research. We should be aware that all three factors need to be accomplished thoroughly to deliver the spoofing attack for radars.

While previous methodology relies on knowledge about the radar bands, researchers are now working on the radar capable of estimating the radar's configuration of target vehicles in runtime and then performing the spoofing attack based on those estimations [4].

On the next level of attacks, the fabricated radar signals are returned to the vehicle's BCW & BCA system. With the weak or absent signal validation mechanism, outcomes could be fake obstacles (false positives) or masked real targets (false negatives).

5.2.2 Compromise of the Camera

The initial step is forgery or simply purchasing necessary physical equipment. The attacker could acquire high-power lasers or specialized high-lumen lights to execute a strong illumination attack. Alternatively, they can buy a projector to deliver an attack based on the projected imagery, especially at night or in a low-visibility environment.

With the required tools in hand, the attacker proceeds to the execution phase, which involves direct interaction and deployment with the environment.

Take adversarial Patches for first example. It requires generating the malicious pattern known as adversarial patch, designed to fool convolutional neural networks. The next critical step is deployment, where the physical patch is attached to a specially chosen target such as a road sign or the roadway itself. When the vehicle's camera captures this patch, the input to the recognition algorithm is already poisoned.

Next example is the strong illumination or laser attacks. Using the previously acquired equipment, the attacker emits a powerful light source or laser beam directly into the camera. This act does not require sophisticated algorithmic knowledge; by overwhelming the physical sensor, it causes temporary blindness, saturation or even permanent damage (which will be discussed in detail in the Denial-of-Service section).

The last vector is the utilization of Projected Imagery, which is the hardest approach to achieve. The attacker uses the projector to cast false visual information into the scene. This could involve projecting fake lane markings, non-existent obstacles, or spoofed traffic signs onto a surface in the vehicle's path. As a passive sensor merely receiving input, the camera captures the fabricated vision with trust and passes it to the image processing algorithms.

5.2.3 Trick the BCW & Mislead the driver

Two attack vectors now intercept, and they both accomplish tricking the BCW system (BCA example is discussed in next attack tree section). As a result, BCW might detect a "fake obstacle" and trigger the warning light and chime. Driver will be confused by the false signals. In SEA scenarios, a fake human figure on the side of vehicle would accidentally lock the back door.

5.3 Denial-of-Service

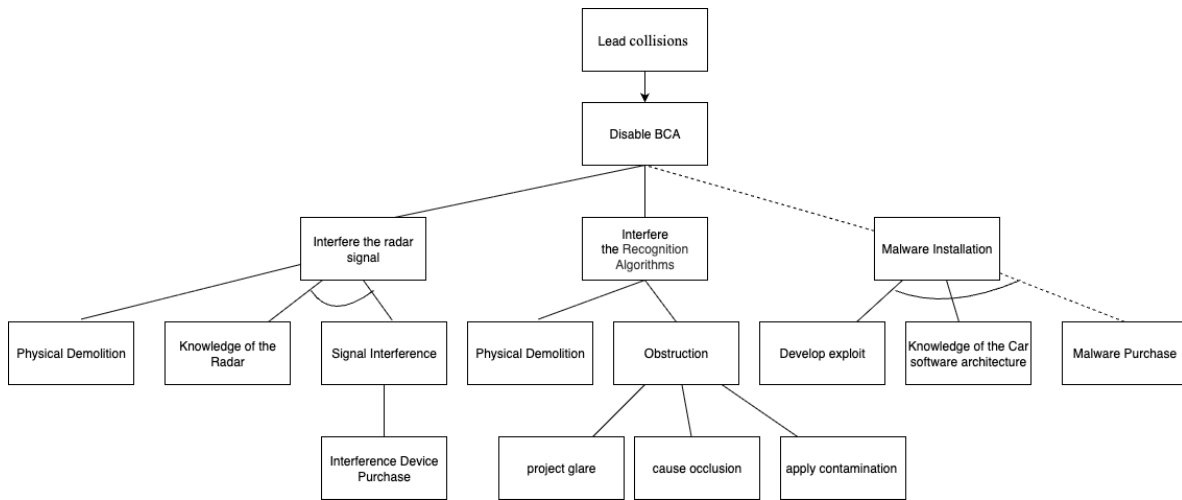


Figure 4: Attack Tree for Denial of Service

The main goal of this attack is to disable the Blind-Spot Collision-Avoidance Assist (BCA) so that the system is no longer reliable to assist the driver, increasing the risk of collisions. The attack tree has divided into two paths, compromising the sensors and the system.

First look at vehicle sensor attacks. In the case of radar, jamming, signal interference[5], or physical destruction are usually used by the attackers. For compromise cameras, attackers normally use occlusion, intense glare projection, lens contamination, or physical damage. Thus, to be successful, there are several preparations needed to attack these sensors. To begin with, the attackers must have access to the basic information about the sensor (the radar parameters or the camera position and lens angle). Knowing that, attackers can select a suitable interference technique and then carry out the attack. For example, signal jamming or injection of electromagnetic waves to radar, or occlusion, glare projection and contamination to cameras.

Next is compromise the system. Software attacks involve installing malware to control or disable BSA without affecting the sensors. At the lowest level, attackers need knowledge of the specific car software architecture. With this knowledge, attackers can either develop an exploit or purchase ready-made malware. Finally, they must obtain physical access to install the malware to achieve the goal.

6 Mitigations

Here we will discuss the mitigation strategies to the three specific threats mentioned above: spoofing, denial of service and elevation of privilege.

6.1 Elevation Of Privilege

Elevation of privilege attack, where adversaries escalate privilege to access engineering mode or modify safety-critical settings, are addressed through setting a privilege separation

control system with least-privilege principle with logging and auditing. Besides, authentication (PIN/physical confirmation) for sensitive auctions is essential for monitoring melilotus behaviors, which would detect and defend potential privilege elevation attack.

6.2 Spoofing

Spoofing attacks involve crafting fake radar echoes or optical imageries to trick the signal processor for BCW & BCA. Considering the attack tree above, here are the three mitigations:

6.2.1 Better encryption for radar waves

To avoid receiving the well-crafted radar signals, Hyundai could add additional layer of security to the radar system, such as time stamps or signatures, making it hard for the attacker to mimic the authentic signals.

6.2.2 Upgrade the optical recognition algorithm

Since the attack vector on the camera side is mainly focused on fooling the convolutional neural network, upgrading the algorithm system may reduce the risk of spoofing attack.

6.2.3 Downgrade to safe mode if spoofing detected

System could downgrade the functionality level if there are signs of spoofing attack take place, for instance receiving inconsistent radar signal (but looks true). By deactivating the BCA or changing “Active Assist” to “Warning Only” mode, we could limit the impact of the spoofing attack, avoiding severe outcomes.

6.3 Denial Of Service

BSD and SEA are strongly related to the radar sensor and camera, So the Denial-of-Service attacks will mainly focus on Those sensors. Considering the attack tree above, here are two mitigations:

6.3.1 Better protection for different interference

To avoid interference such as electromagnetic waves, jamming noise, or occlusion, Hyundai could add protection in two aspects. In terms of hardware, simply notifying the driver of occlusion is not enough. The sensor needs both a self-cleaning system to ensure it works properly while driving and protection against electromagnetic waves. In terms of software, upgrading the algorithm to resist jamming noise is also important.

6.3.2 Securing Debug Ports in Automotive Systems

In in-car systems, debug ports are usually ignored because most people never notice them. Hiding the debug port is not enough. If an attacker gains access to a debug port, they can install malware that causes a denial of service or, even worse, covertly monitor the driver. The port should only function within a certified area, and any connecting device should also be certified.

7 Summary

Under the offline model assumption, three threats can be identified at the margins of two trusted boundaries. If the Blue Link system, involving cloud and Internet interactions, is considered, the threat model becomes substantially more complex and demands additional layers of analysis and mitigation.

In the future, we plan to expand this analysis to online and cloud-connected attack surfaces, such as the Blue Link system, to address the broader cybersecurity issues of modern vehicles.

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